

Patient Follow-up Agent

Proposal for a pilot deployment — autonomous dental recall management

Prepared for	Clinic operations and IT decision-makers
Scope	Pilot: automated patient recall, delivery and escalation
Deployment revision	fbcc6c1ca09a7624fd5f01016e674cacdc4a5287
Branch	master — “Merge pull request #3 from pioneer6-ai/feature/patient-portal”
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1. Executive summary

Dental clinics lose revenue and patient relationships to a problem that is administrative, not clinical: recall. Patients fall out of the six-month cycle, nobody notices until the file is opened, and by then the case is months stale.

This proposal puts an agent in charge of that cycle. It watches the patient list, decides who needs contact and how urgently, sends the reminder over the best available channel, reads the outcome, and escalates to a human when the machine should stop. Every decision and every delivery attempt is written to an auditable record.

The system is **already built and running**. It is not a concept: the code is committed at the revision named above, 811 automated tests pass in twelve seconds, the operator dashboard and a patient-facing portal both serve live data, and the AWS messaging paths are wired to Amazon SES and AWS End User Messaging SMS. Every claim in this document is traceable to an artifact in the evidence pack rather than to a demonstration.

The remaining work is not development. It is provisioning: moving the AWS account out of the SES sandbox, onboarding the clinic’s own domain so reminders reach the inbox consistently rather than landing in the spam folder, and subscribing the SMS channel.

Two findings from the evidence deserve to be read before anything else in this document, because both were discovered by measuring rather than assumed:

1. **Where a live message lands is not yet under our control.** The provider counted one message as delivered while the mailbox filed it in the spam folder (section 7, evidence 7); three later reminders from the same identity reached the inbox unmarked (evidence 8). A reminder nobody reads has not reminded anyone — and at present we cannot predict which of the two outcomes a given send will produce.
2. **Once a case is escalated, the agent stops touching it.** That is the correct design — clinical ambiguity belongs to a human — but it means the case is not re-raised if the alert is missed. In the measured run, 9 of 11 cases ended in that state (section 9).

Neither is a reason to withhold the system. Both are reasons to sequence the rollout as section 11 describes.

2. The problem

A recall programme fails in three predictable ways.

It depends on someone remembering. Recall is a background task. It is the first thing dropped when the front desk is busy, and it is invisible until a patient mentions they were never contacted.

It does not know when to stop or when to escalate. A reminder that is sent seven times is harassment; a complex case that is reminded automatically is a clinical risk. A workable system needs a defensible rule for both.

It cannot be audited after the fact. “Did we contact this patient?” is a question clinical governance will ask. If the answer lives in an inbox and in someone’s memory, it cannot be answered.

Any automated recall system has to fix all three. Automation that is not auditable is worse than manual work, because it hides its own failures.

3. What is being proposed

A single agentic service that owns the recall cycle end to end.

Capability	Behaviour
Watch the patient list	Retrieves active patients, identifies those overdue against clinic policy, and filters out the ones not yet due
Prioritise	Scores urgency from treatment type, elapsed time and patient history, using weights the clinic can edit
Choose action per case	Remind, leave alone, or escalate — chosen only from the actions the rules permit
Send across channels	SMS, email and WhatsApp, with automatic fallback when a channel is unusable
Perceive failure	Reads <i>why</i> a send failed and routes around it instead of reporting a false success
Escalate	Hands the case to staff with a reason once the reminder budget is spent or every channel has failed
Record everything	Durable audit trail of every decision, attempt, outcome and patient reply
Show the operator	Web dashboard: cases, escalations, statistics, audit log, and a scheduling calendar
Let the patient self-serve	Tokenised patient portal: status, available slots, booking, and a question channel

The clinical logic is deliberately boring, and that is the point. Which reminders are safe to send is decided by a deterministic rule engine from the clinic’s own policy. Time, treatment type and history produce a score a clinician can read and challenge.

4. How it works

4.1 The loop

The agent runs a five-phase cycle rather than firing one-off messages.

- 1. **Trigger** — a deterministic filter (`core/trigger_service.py:64`) drops opted-out patients and cases not yet due for their explicit follow-up date, before any reasoning is spent on them.
- 2. **Perceive** — load active patients, detect who is overdue, and pick up the outcome of previous attempts.
- 3. **Decide** — compute what is permissible for each case and pick one action.
- 4. **Act** — send on the best channel, book appointments, or escalate to staff.
- 5. **Observe** — read replies, update case state, and write the audit record.

Failure is an input, not an exception. A send that comes back as `recipient_not_verified` or `provider_unavailable` is fed straight back into Perceive, and the agent tries the patient’s next reachable channel.

4.2 Where judgement sits, and where it does not

Three questions are kept apart on purpose.

Question	Decided by	Why
Which actions are <i>safe</i> for this case?	Deterministic rule engine from clinic policy	Clinical safety must not depend on a model being reachable or correct
Is this action <i>authorised</i> ?	PolicyGuard , a deterministic authorization layer (<code>agent/policy_guard.py:130</code>)	Consent, opt-out and emergency handling must be arithmetic, not inference
Which permissible action to take <i>now</i> ?	The model, via a tool call	Tone, timing and when to involve a person are judgement calls

The model **cannot** take an action the rules did not offer. If it is unreachable, or answers out of bounds, the rules decide instead — the agent degrades, it does not stop. The dashboard builds its decision engine through `FollowUpAgentOrchestrator.with_llm_decisions` (`agent/orchestrator.py:383` , `web/app.py:100`), which resolves the vendor from `AGENT_LLM_PROVIDER` via `LlmDecisionEngine.from_environment` (`agent/decision.py:384`) and falls back to the rule engine when no usable model is configured.

The measured run in section 9 is nevertheless a **rules-only** run, and this is worth being precise about: it is rules-only because the evidence harness builds the orchestrator directly (`proposal/scripts/generate_audit_record.py:94`) and so pins the deterministic rule engine – which is

what makes the recorded counts reproducible by a reviewer – and not because a model was missing. Every one of that snapshot’s 462 decisions reads `Decided by: rules`. The model-backed path is wired, installed and separately exercised; section 9.4 records that trial and says plainly what it does and does not prove.

`PolicyGuard` sits between the two and is checked before every action. Its rules are ordered and first-match-wins:

	Order	Rule	Effect
	1	<code>action_allowlist / noop_allow</code>	An unrecognised action is never authorised
	2	<code>emergency_override</code>	A medical emergency outranks everything else
	3	<code>opt_out_record / opt_out_enforced</code>	An opt-out is recorded and then binding
	4	<code>clinical_question_boundary</code>	Clinical, financial and administrative questions are handed to staff, not answered by the agent
	5	<code>booking_consent_clear / _ambiguous / _missing</code>	Booking requires unambiguous affirmative consent
	6	<code>default_allow</code>	Otherwise the action stands

Which engine chose is recorded per decision, so the distinction survives into the audit trail.

4.3 When a send fails

This is the behaviour the design exists for. Message sending returns a structured outcome, never an exception:

Situation	Outcome	What the agent does
Recipient not verified / not allowed	<code>recipient_not_verified</code>	Try the patient’s next reachable channel; escalate if none work
No contact detail for that channel	<code>missing_recipient</code>	Skip the channel, try the next
No automated sender exists	<code>unsupported_channel</code>	Try the next channel
Account not onboarded to the service	<code>not_subscribed</code>	Try the next channel, then escalate — a human must fix this
Provider or network problem	<code>provider_unavailable / network_error</code>	Try the next channel
Empty body or subject	<code>invalid_request</code>	Treat as a defect: record it, do not retry forever

The agent never reports a reminder as sent when it was not, and never abandons a case: once the reminder budget is spent or every channel has failed, the case is escalated and staff are alerted.

4.4 The live-send gate

Real transmission requires two independent switches:

```
MESSAGING_DRY_RUN=0    # tools layer: allow real transmission
AGENT_LIVE_SENDS=1     # agent layer: and the agent means it too
```

With either missing, the agent renders what it *would* have sent instead of sending. The reasoning is that a fully populated configuration file must never be enough on its own to message a patient. During evaluation this gate is the reason the system is safe to run against realistic data — including the evidence capture in evidence 1, which asserts at both ends that the offline printer was the only backend constructed.

4.5 The clinic’s clock is an input, not a call to `now()`

Every date decision in the deployed revision comes from an injectable clock (`core/clock.py`), so “today” is never hardcoded. Production runs on `SystemClock` bound to the clinic’s timezone; tests and the evidence capture run on `FixedClock` and replay a whole quarter in seconds. This is what makes the behaviour in section 9 reproducible rather than anecdotal.

5. Delivery architecture

5.1 Channels

Channel	Transport	Status
SMS	AWS End User Messaging SMS (<code>pinpoint-sms-voice-v2</code>), transactional message type	Wired in code (<code>tools/aws_providers.py:307</code>); the account is not subscribed to the service , so sends return <code>not_subscribed</code> today
Email	Amazon SES (<code>send_email</code>) from a verified identity (<code>tools/aws_providers.py:348</code>)	Wired and verified delivering (evidences 6 and 7)
WhatsApp	Meta Cloud API / Twilio	Wired, not configured for this clinic
Phone call	No automated sender	Detected as unsupported, escalated instead

Each channel is a separate provider behind one interface, so what a channel can do — and what it cannot — is visible in one place. During the pilot the account is intentionally restricted: SMS and email refuse any recipient outside an explicit allow-list of verified test contacts, returning the same `recipient_not_verified` shape the agent already knows how to fall back from.

The channel tools are the agent’s own tool interface, registered with Claude’s tool-use schema so the model calls them the same way it calls any other action: `send_sms(phone_number, message, reason)` at `tools/messaging.py:449` and `send_email(to_email, subject, body, reason)` at

`tools/messaging.py:482` . Both take a mandatory `reason` , which is printed to the operator log as `[Agent决策] ...` and stored, so the agent’s justification for contacting a patient is recoverable from the log as well as the audit record.

5.2 Bringing the clinic’s own identity

Two integrations make the pilot look like the clinic rather than like a vendor.

The clinic’s own model. The decision engine accepts any OpenAI-compatible provider — Claude, OpenAI, Azure, a self-hosted service inside the clinic network. Swapping providers is configuration, not code.

The clinic’s own mailbox. Email can be sent from the clinic’s existing domain address, so messages are signed by the clinic’s mail provider and inherit its sending reputation. An alternative SMTP path is available where SES is not appropriate.

This pilot now sends under a clinic-named identity rather than a developer’s personal address: `brightsmileagent@gmail.com` , verified in `ap-southeast-1` , with `+6585144321` configured as the SMS origination identity (evidence 2). Both email transports carry it — the SES tool sends from that address, and the SMTP path the agent’s own email channel uses is configured with it as both the authenticated mailbox and the From address, so patients see `BrightSmile Dental <brightsmileagent@gmail.com>` either way. Naming the sender after the clinic is a presentation matter and does not confer authentication — a `gmail.com` address cannot carry a clinic-controlled SPF, DKIM or DMARC record, so it borrows Google’s reputation instead of earning the clinic’s. The move to an authenticated clinic-owned domain remains the rollout gate described in section 12.

5.3 Proving delivery, not just acceptance

A provider message ID only proves the message was *accepted*. Amazon SES is therefore configured with a configuration set and event destination, so every send emits per-message `Delivery` , `Bounce` and `Complaint` events, readable in CloudWatch under the `AWS/SES` namespace. Delivery is a separate claim from transmission and is evidenced separately.

This was not left as theory. A live message was sent through the agent’s own tool layer and then traced from both ends — evidence 6 records the provider’s acceptance, evidence 7 records the provider’s `Delivery` telemetry for that message *and* an independent search of the recipient mailbox.

Check	Source	Result
Provider acceptance	SES <code>send_email</code> response	Accepted, message id <code>010e01a0d7e70489-...-000000</code>
Provider delivery	CloudWatch <code>AWS/SES</code> , config set <code>patient-followup</code>	Delivery 6, Bounce 0, Complaint 0
Mailbox arrival	IMAP search of the recipient account	Found — in spam

That trace surfaced one of the two operational facts that matter most in this pack: **the message was delivered, but the mailbox filed it as spam**. It arrived, and it would not have reminded anyone. That is not a defect in the agent; it is what happens when a sender identity has no domain authentication behind it. It is the concrete reason the rollout plan insists on the clinic’s own authenticated domain rather than a working default.

It is not, however, a uniform outcome, and evidence 8 is where that becomes visible: reminders sent from this same identity five hours later arrived in the **inbox**, unmarked, with the agent's per-patient wording intact. Placement is presently inconsistent rather than uniformly bad, which is a weaker claim than "the mail is blocked" and a harder one to engineer around. It makes verified inbox placement a rollout gate rather than an assumption, because the sending identity's reputation — not the message — is what decides.

6. What this revision adds

The revision under evaluation is not the same system that was assessed earlier. It carries a full patient-facing surface and a set of safety and configuration controls that did not previously exist. All of it is in the evidence pack.

Addition	Where	Why it matters
Patient portal	<code>web/app.py:933</code> , <code>/api/patient-portal/<token>/*</code>	Patients see their status, view available slots, book, decline, or ask a question — through a per-patient token, not a login
Patient chat assistant	<code>agent/patient_chat.py:135</code>	Classifies patient questions and routes clinical/financial ones to staff rather than answering them
Deterministic authorization	<code>agent/policy_guard.py:130</code>	Consent, opt-out and emergency rules evaluated before any action, with the rule name recorded
Editable urgency policy	<code>core/urgency_config.py:11</code> , <code>web/app.py:1306</code>	Clinic can change urgency weights and thresholds without a code change
Change preview	<code>core/preview_engine.py:86</code> , <code>web/app.py:1367</code>	See what a policy change <i>would</i> do to open cases before applying it
Trigger service	<code>core/trigger_service.py:64</code>	Explicit “is this case due today” filter, separate from scoring
Slot ranking	<code>core/slot_ranking.py:72</code>	Offers appointment times ranked by proximity to the missed appointment session
Scheduling integration	<code>core/scheduling_calendar_adapter.py:114</code> , <code>scheduling/</code>	Real calendar with blocked periods, persisted bookings and slot capacity
Injectable clock	<code>core/clock.py</code>	Deterministic date behaviour, testable and replayable
Bulk patient import	<code>web/app.py:628</code> , <code>web/app.py:668</code>	CSV and Excel upload, with an escalated-case import path
Staff login and calendar	<code>web/app.py:159</code> , <code>web/app.py:182</code>	Operator authentication and a working day view

The portal is not a screenshot. Evidence 5 minted a real portal link through `POST /api/patients/<id>/portal-link` against the running application and rendered the resulting page, so the capture is of that process’s own output.

7. Deployment status

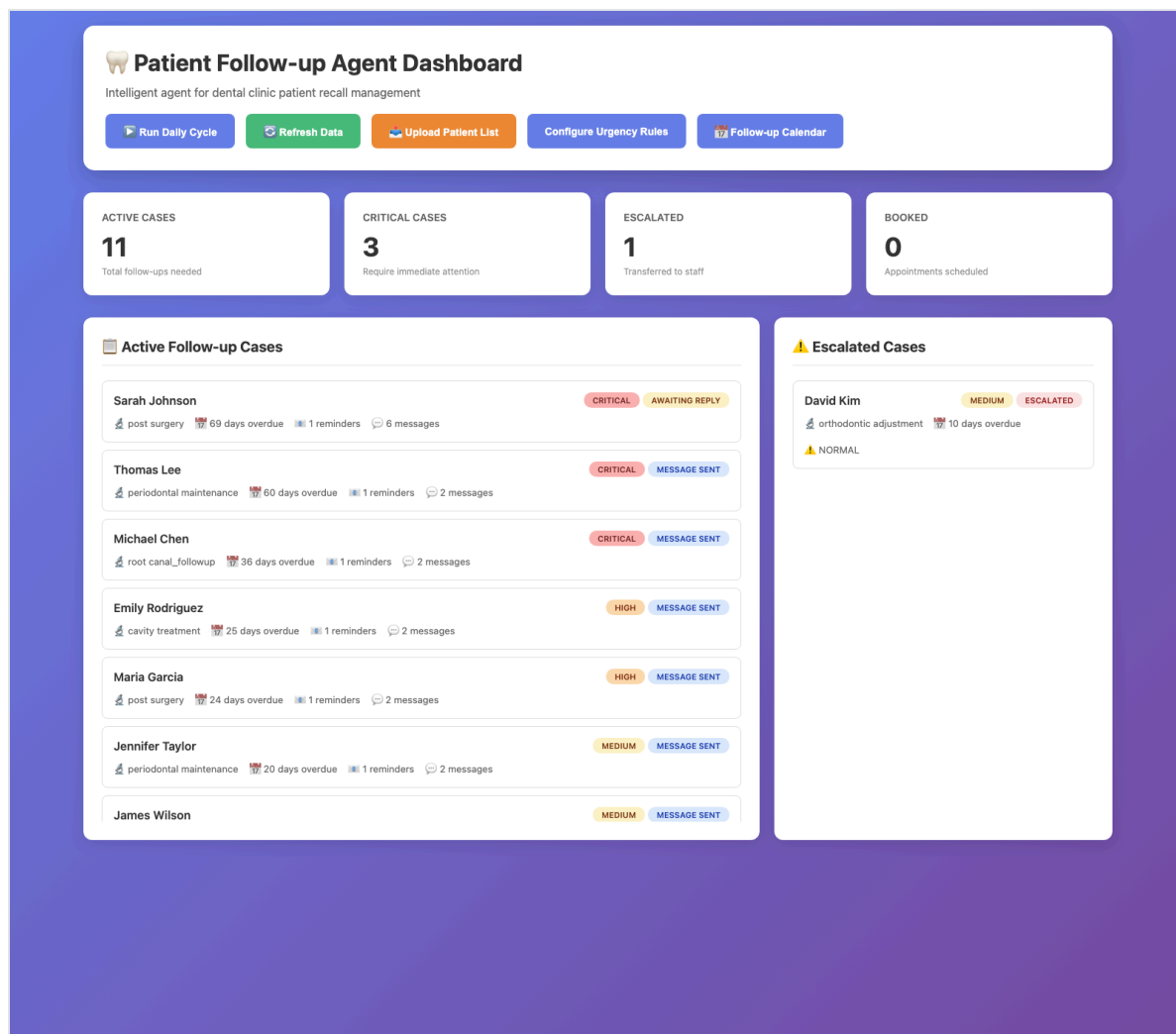
7.1 Running and evidenced today

- The application runs and serves its dashboard, its patient portal and its JSON API, and accepts operator commands: the capture drove a full daily cycle over the sample patients and three reply scenarios, producing 11 cases in five distinct states — evidence 5.
- The full automated suite passes on a pinned revision and interpreter — evidences 3 and 4. **811 tests across 30 test modules, 12,119 lines of test code, in 12.35 seconds.**
- The recall loop was replayed over 42 simulated days and recorded **493 events: 463 decisions and 30 communication attempts across 11 patients** — evidence 1. Note the scope qualification in section 9.3.
- The audit record is machine-readable and the counts above are recomputed from a hash-identified snapshot rather than quoted from a summary.
- The AWS transport configuration is pinned to named services, a named region, explicit sender identities and an explicit recipient allow-list — evidence 2.
- A real message was sent through the tool layer, accepted by the provider, counted as delivered in the provider’s own telemetry, and found in the recipient mailbox — evidences 6 and 7.

The deployed codebase (evidence 3, 78 files and 32,073 lines):

Area	Files	Lines
tests/	30	12,119
tools/ (messaging, AWS, LLM, transport)	13	6,118
agent/ (orchestrator, decision, delivery, audit)	9	4,635
web/ (dashboard, portal, APIs)	5	3,477
core/ (models, clock, policy, scheduling)	11	1,978
scheduling/	4	1,321
utils/	2	1,022
root (app , demo)	2	1,115
scripts/	2	288
Total	78	32,073

Validated on CPython 3.14.5 with boto3 1.35.36, botocore 1.35.99, flask 3.0.0, openpyxl 3.1.2 and certifi 2026.7.22.



The operator dashboard, populated with the project's built-in sample data

The screenshot is not of an idle page: the capture process started the real application, ran the agent loop through the dashboard's own control surface, and then rendered the result. The board shown — 11 cases in five states, urgency weighted critical 3 / medium 3 / low 3 / high 2 — is what that run produced. The capture runs in a scratch directory so that exercising the agent here cannot append to the audit trail evidence 1 is derived from.

Hi Sarah Johnson 🍌

Pick a time that works for you below.

awaiting reply

Available appointment times

9月28日 星期一

9:00 AM	9:30 AM	10:00 AM	10:30 AM	11:00 AM	
11:30 AM	1:00 PM	1:30 PM	2:00 PM	2:30 PM	3:00 PM
3:30 PM	4:00 PM	4:30 PM			

9月29日 星期二

9:00 AM	9:30 AM	10:00 AM	10:30 AM	11:00 AM	
11:30 AM	1:00 PM	1:30 PM	2:00 PM	2:30 PM	3:00 PM
3:30 PM	4:00 PM	4:30 PM			

9月30日 星期三

9:00 AM	9:30 AM	10:00 AM	10:30 AM	11:00 AM	
11:30 AM	1:00 PM	1:30 PM	2:00 PM	2:30 PM	3:00 PM
3:30 PM	4:00 PM	4:30 PM			

10月1日 星期四

9:00 AM	9:30 AM	10:00 AM	10:30 AM	11:00 AM	
11:30 AM	1:00 PM	1:30 PM	2:00 PM	2:30 PM	3:00 PM
3:30 PM	4:00 PM	4:30 PM			

10月2日 星期五

9:00 AM	9:30 AM	10:00 AM	10:30 AM	11:00 AM	
11:30 AM	1:00 PM	1:30 PM	2:00 PM	2:30 PM	3:00 PM
3:30 PM	4:00 PM	4:30 PM			

None of these times work — remind me next week

Questions? Ask our assistant

Ask about your appointment or general dental care... Send

The patient portal, rendered from a token minted by the running application

7.2 Configured but not yet proven live

- **SES production access.** The account is in the SES sandbox: 200 messages per 24 hours at one message per second, and senders and recipients must both be verified. This is appropriate for a pilot test population and unsuitable for a real one.
- **SMS is not provisioned.** Reading the account state returns `SubscriptionRequiredException` for End User Messaging SMS in every region checked, and no verified destination numbers are visible. The SMS code path is complete and its failure handling is exercised, but the service itself is not subscribed, so SMS attempts currently return `not_subscribed` and fall through to the next channel. **This is a provisioning task, not a development one.**
- **Deliverability.** Placement is currently inconsistent: one message was filed in the recipient's spam folder (evidence 7) while three later ones reached the inbox (evidence 8). Sending must move to a clinic-owned domain with SPF, DKIM and DMARC configured, and verified inbox placement must be demonstrated, before reminders can be trusted to reach a patient.
- **The model-backed decision path.** This pack's decision counts come from the rule engine because the evidence harness pins it (`proposal/scripts/generate_audit_record.py:94`), not because a model was unavailable: the SDK is installed, a vendor is configured, and the dashboard wires the model

(`agent/orchestrator.py:383` , `web/app.py:100`). What the pack does not contain is a *pinned, hash-identified* artifact for that path. The trial in section 9.4 is observed behaviour in a running deployment, and this document does not dress it up as more than that.

Section 13 lists the decisions required to close these.

8. Evidence pack

Every artifact lives under `proposal/evidence/` and was generated by the scripts in `proposal/scripts/` against the revision named on the first page.

#	Artifact	What it proves
1	01_agent_decision_audit.md	The recall loop ran 42 simulated days, making 463 decisions and recording 30 communication attempts, from a hash-identified snapshot
2	02_aws_messaging_config.md	The AWS transport is configured against named services, region, sender identities and allow-list, with the code paths cited by file and line
3	03_runtime_and_repository.md	The exact revision, interpreter, dependency versions and codebase size every other claim was validated on
4	04_test_suite.md	811 automated tests pass, with verbatim pytest output retained
5	05_dashboard_runtime.md	The application starts, serves its dashboard and patient portal, answers its JSON API and executes operator commands
6	06_live_delivery.md	A real message was transmitted through the agent's own tool layer, and the provider accepted it with a message id
7	07_delivery_confirmation.md	That message was counted as <i>delivered</i> by the provider's own telemetry, and independently found in the recipient mailbox — where it had been filed as spam
8	08_patient_inbox_receipt.md	Three later reminders from the same identity reached the recipient's inbox , unmarked, each carrying the per-patient wording the agent generated
—	artifacts/	Raw primary sources (audit snapshot, generation sidecar, pytest output) so figures can be re-derived rather than trusted
—	screenshots/	The rendered dashboard, the patient portal, and the recipient's inbox as the patient saw it
—	00_index.md	Generated index of every artifact above with its hash and stated purpose

Reproduction is a single command per artifact; the appendix lists them. Report 8’s image is the one exception — it was contributed by the operator rather than captured, and the index marks it as such rather than implying a command that does not exist.

The same eight reports are also consolidated into a single printable document, `proposal/Patient_Followup_Agent_Deployment_Evidence.pdf`, which restates every figure, re-checks each artifact hash and states plainly what the pack does and does not prove. It is generated from `proposal/deployment_evidence.md` by `proposal/scripts/build_deployment_evidence_pdf.py`, so it cannot drift from the pack without the build being re-run. Read it when the reports need to be read together; read the reports themselves when a single claim is in question.

9. Measured behaviour

Computed from the audit snapshot identified in evidence 1. The snapshot was produced by driving the real orchestrator with a fixed clock from **21 September 2026 to 1 November 2026 — 42 daily cycles over 11 open cases** — with five patient replies injected so the record covers the Observe path and not only outbound decision-making.

Decisions (463)

Action	Count	Urgency at decision time	Count
do_nothing	430	critical	209
send_reminder	22	high	150
escalate_to_staff	11	medium	78
—	—	low	26

Communication attempts (30; 26 outbound, 4 inbound patient replies)

Channel	Attempts	Succeeded	Failed	Success rate
email	12	12	0	100.0%
sms	9	9	0	100.0%
whatsapp	7	7	0	100.0%
phone_call	2	2	0	100.0%
Overall	30	30	0	100.0%

9.1 The success rate is not a deliverability claim

Every attempt is recorded as successful because the delivery backend in this run is the offline printer: it accepts any recipient it is handed and therefore cannot fail. The offline mode was asserted at both ends of the capture — every constructed channel had to be a `PrintDeliveryBackend` (`agent/delivery.py:169`), and

the transmitting backend's `deliver` was patched to raise. The 100% figure is therefore a property of the offline backend, **not** of carriers, and the absence of failures here is **not** evidence that the fallback path works. Carrier-level delivery is evidenced separately, for one real send, by evidences 6 and 7.

An earlier draft of this section described the failures as “the fallback working as designed”. That sentence was removed because the measured run produced no failures to read: it would have been a claim about a code path the evidence does not exercise.

9.2 Escalation is real, and terminal

- 11 escalation actions, touching **9 of 11 distinct patients**.
- Escalations are 2.4% of all decisions, but that understates them: measured against the 33 decisions that actually chose to contact or hand over a case, escalation is 33%.
- **Once a case is escalated it is closed to the agent.** A terminal status yields `do_nothing` forever (`agent/decision.py:187`). The audit record shows this plainly: of the 430 `do_nothing` decisions, **281 were taken on cases already in the escalated state**. Final case statuses after 42 days: 9 escalated, 1 declined, 1 booked.

That is the design working — clinical judgement stops the machine — but it has an operational consequence the clinic must accept: **the case is not re-raised**. If the escalation alert is missed, nothing else will surface the patient. A reply *does* reopen the case (one injected reply moved `escalated` to `awaiting_reply`), so the patient is not locked out; a silent patient is. The mitigation is in the rollout plan: escalation alerts must land somewhere that is actually monitored, and this is a decision for the clinic rather than a configuration change.

9.3 Scope qualification — important

This deployment's *agent layer* is not configured for live transmission (the second switch, `AGENT_LIVE_SENDS`, is not set). The attempts above therefore evidence the full decision, routing, fallback, escalation and audit pipeline, but they are not, by themselves, proof of transmission to patient handsets and mailboxes. That proof is supplied separately, for one real send, by evidences 6 and 7, and those are the only artifacts in this pack that claim transmission rather than behaviour.

One further qualification on the record itself: the audit logger timestamps entries with its own wall clock, not the injected clock. Under the fixed-clock harness the entry timestamps therefore show a sub-second window while the decisions they describe span the simulated quarter. In production the two are the same thing; in this pack the simulated period is the meaningful span. This is noted in the evidence artifact rather than hidden.

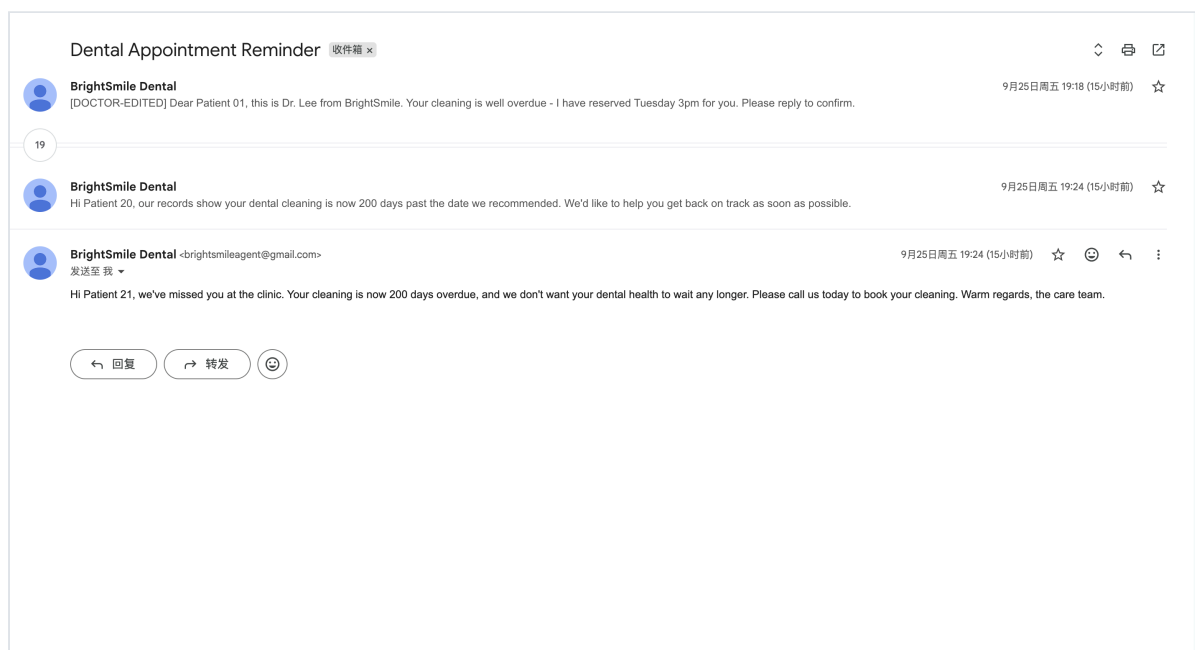
9.4 The model-backed path, exercised

Section 9 above is a rules-only record by construction, and section 7.2 of an earlier draft read that as the model path being unexercised. That inference was wrong, so this subsection records the model path separately, and bounds the claim to what was actually observed.

What was run. The dashboard (`web/app.py`), with a model configured for both decisioning and message authoring, against a cohort of 21 patients all scored `critical`. The agent drafted a reminder for each and queued it. A staff member edited one draft in the review panel; the panel's “Confirm Selected” then claimed the batch and delivered it: 21 sent, 0 failed, queue empty.

What it establishes.

- *The model decides.* The live audit record holds 944 decisions stamped `Decided by: llm`, alongside 11 `llm-error` and one `llm-guardrail` rejection. The fallback taxonomy described in section 4.2 therefore fires under real conditions, and every degradation carries its own reason rather than being swallowed.
- *The model authors the body, not only the action.* The generated prose differs from the deterministic template, so authorship is verifiable rather than assumed.
- *A staff edit is honoured.* The text the audit record shows as transmitted is the edited text, not the draft it replaced.
- *The confirmation gate holds.* Drafting transmitted nothing, and a cancelled batch of 21 recorded no outbound events at all.
- *Some of it reached an inbox.* Three of the batch’s messages were seen in the recipient’s inbox, unmarked, five hours after the trial (evidence 8). The recipients’ side of the batch is therefore documented, not only the sender’s.



The recipient’s mailbox, showing three of the batch’s reminders delivered to the inbox

The screenshot is the recipient’s view of the same batch the console reported as `21 sent, 0 failed`: one thread, three messages, every sender line reading BrightSmile Dental. The three bodies differ from one another in greeting, phrasing and closing, and none of those sentences appears in any template in the repository — that is the model authorship the audit record claims, seen from the receiving end. The message marked `[DOCTOR-EDITED]` is the staff edit, and it is worth noting that the marker itself is not a label the software emits: it is the operator’s own text, carried through the draft-edit path (`agent/orchestrator.py`, `update_pending_send`). The image cannot be re-derived and its recipients are sample records, so evidence 8 states both limits rather than leaving them to be inferred.

What it does not establish.

- *Not deliverability.* All 21 sends were addressed to a single verified test mailbox rather than to 21 separate patients, and the SMS channel is unprovisioned in this account (section 7.2). The trial exercised the email path only. Evidence 8 shows three of those messages in the inbox, while the spam-placement finding in section 7.2 shows an earlier message in spam: the two together mean placement is inconsistent, not that it is solved. This remains the strongest reason the rollout plan gates on a clinic-owned authenticated domain.

- *Not a placement rate.* Three messages in one mailbox is an existence proof of inbox delivery. It says nothing about what fraction of a real cohort would arrive, and evidence 8 states that limit itself.
 - *Not a frozen count.* The figures above are quoted as at the observation on 25 September 2026. The record is append-only and still being written to: the same log now carries 1,600 `Decided by: llm` decisions, while the 11 `llm-error` and one `llm-guardrail` entries are unchanged. Read these numbers as a snapshot of a moving log, which is why the log itself is not shipped as a fixed artifact.
 - *Not a reproducible artifact.* This is observed behaviour in a running deployment on 25 September 2026, read from the live append-only audit log. Neither the audit log nor evidence 8's screenshot is offered as a hash-identified, script-reproducible member of the evidence pack in section 8 — the log because it is still being written to, the screenshot because it was contributed by hand — and neither should be cited as though it were.
-

10. Safety, compliance and auditability

Consent and contactability. Every send is attempted only on a channel the patient record marks as reachable. A patient who cannot be reached on any channel is escalated to staff rather than being contacted some other way.

Authorization is deterministic. `PolicyGuard` evaluates every proposed action against ordered rules before it executes, and records which rule decided (`agent/policy_guard.py:156`). A model cannot authorise its own action; the worst a bad model output can do is choose among actions the rules already permitted, and an out-of-bounds choice is refused.

Restraint is a setting, not a habit. Maximum reminder count, follow-up interval and urgency weights all come from clinic policy, editable at runtime through the dashboard with a preview of the effect before it is applied. The clinic can dial the system's eagerness up or down without a code change.

Auditability. Every decision records the case, the action, the urgency and case status at decision time, the justification, and which engine chose. Every communication records the channel, direction and outcome. Communication justifications are printed as `[Agent决策] ...` in the operator log and stored in the record, so the reason for contacting a patient is always recoverable.

Human authority. Escalation is a first-class action, not an error path. The agent is designed to hand difficult, unresponsive or clinically ambiguous cases to people — including any clinical, financial or administrative question a patient asks through the portal, which is routed to staff rather than answered.

Data handling. The dashboard, the portal and the evidence pack in this document are populated with the project's built-in sample data, not patient records, and the evidence pack contains no credentials. Sending in the pilot is restricted to an explicit allow-list of verified test contacts.

11. Proposed rollout

Phase	Duration	Content	Exit criterion
1. Pilot on test contacts	1 week	Run the loop against a small allow-list of clinic-owned test contacts; validate copy, tone, timing and escalation	Signed-off message templates and escalation thresholds
2. Production provisioning	1–2 weeks	SES production access, clinic-owned domain with SPF/DKIM/DMARC, SMS service subscription and origination identity	The sandbox limits in section 7.2 no longer apply, and repeated test sends reach the inbox consistently rather than inconsistently
3. Supervised cohort	2–4 weeks	Real patients, restricted cohort; staff review every escalation; messages capped well below policy maximums	Escalation rate and patient responses match clinical expectation
4. General rollout	Ongoing	Full patient population within policy limits, with monthly review of the audit record	—

Two sequencing points matter more than the schedule. **Deliverability gates phase 3**, not phase 2: until a clinic-owned authenticated domain is sending, where a reminder lands is not something we control — one message in spam and three in the inbox from the same identity is the measurement that says so — and enrolling real patients while that is unresolved is testing the wrong thing. **Escalation monitoring gates phase 3** for the reason in section 9.2 — a terminal case is not re-raised, so the alert channel has to be one staff actually watch.

Each phase is reversible at the switch level: disabling either live-send gate returns the system to rendering rather than sending, with no data migration.

12. Risks and mitigations

Risk	Impact	Mitigation
A reminder is sent to the wrong patient	Clinical and reputational	Rule engine gates every action; PolicyGuard authorises it; allow-list during pilot; validation against the same permitted-action list for model choices
Provider outage	Delay in recall	Per-channel fallback, then escalation; the case is never dropped
Duplicate or excessive contact	Patient annoyance, complaints	Reminder budget from clinic policy applied before escalation; quiet periods enforced as a hard floor
Sandbox limits mistaken for production capacity	Under-delivery at scale	Stated explicitly in section 7.2; quota is checked as part of the evidence pack
Silent delivery failure	False confidence	Provider ID treated as acceptance only; SES delivery events configured as the separate delivery claim; the live check in evidence 7 caught a real deliverability failure that a message id alone would have hidden
Sending identity has no domain authentication, so placement is unpredictable	Recall silently fails while reports show success	Measured, not hypothetical — evidence 7 found one message in spam, evidence 8 found three later ones in the inbox. Not deterministic, which is worse to plan around than a consistent failure. Mitigated by moving to a clinic-owned domain with SPF, DKIM and DMARC before any real patient cohort, and by treating verified inbox placement as a rollout gate rather than an assumption
An escalation is missed and the case is never re-raised	A patient in difficulty goes uncontacted	Measured, not hypothetical — 9 of 11 cases ended escalated, and a terminal case is not revisited (section 9.2). Mitigated by routing escalations to a monitored queue, applying an ageing report to the escalation list, and confirming the monitor with the clinic before phase 3
SMS channel unavailable at the account level	Reminders lean on email only	<code>not_subscribed</code> is already a known outcome the agent falls back from; provisioning the service and an origination identity is a tracked open item
Test activity contaminating the audit record	Weakened governance	Found during this work: the audit logger resolves <code>audit_log.json</code> relative to the working directory and

Risk	Impact	Mitigation
		exposes no injectable path (<code>agent/action_handlers.py:305</code>), so the test suite appends to the production record — in this workspace the file grew from 62,151 to 124,306 bytes purely from running pytest. Evidence 1 uses a frozen hash-identified snapshot to stay reproducible, and the dashboard capture runs in a scratch directory for the same reason. Isolating the audit path in the product itself is a prerequisite for production sign-off
Model unavailability or unsafe output	Broken decisioning	Rules-only path is the fallback; behaviour is identical, minus judgement calls. Exercised in the live trial (section 9.4): against the 944 accepted model decisions observed at that time, the record shows 11 <code>llm-error</code> and 1 <code>llm-guardrail</code> fallback, each carrying its own reason rather than degrading silently

13. Open items requiring a decision

1. **Request SES production access** for the clinic's sending identity.
2. **Move sending to a clinic-owned domain** with SPF, DKIM and DMARC, and treat inbox placement — not provider acceptance — as the rollout gate. Evidence 7 shows why this is the highest-value item in the list.
3. **Subscribe to and onboard the SMS service.** The origination identity (`+6585144321`) is already chosen and configured, but the account is not subscribed, so no SMS can be sent at all. This is the only item here that carries a per-message cost, and it is the one capability in this proposal that cannot be demonstrated even partially until it is done.
4. **Confirm where escalation alerts land, and that the destination is monitored.** Section 9.2 makes this a clinical decision, not a technical one.
5. **Confirm clinic policy values** for reminder budget, follow-up interval and escalation recipients.
6. **Decide the audit retention policy** and, before production, make the audit path injectable so test runs and demonstration runs cannot write to the production record. The evidence pack works around this today; the product should not have to.
7. **Confirm which model the clinic will run in production, and sign off on its cost, data residency and vendor.** This is no longer a build task: a provider is configured, the SDK is installed, and the dashboard uses it (section 9.4). Whichever engine is chosen should be stated in the deployed revision rather than left to silently degrade.

Appendix: reproducing the evidence

Run from the project root, after activating the virtual environment:

```
.venv/bin/python proposal/scripts/generate_audit_record.py      # regenerates
  audit_log.json
.venv/bin/python proposal/scripts/capture_audit_evidence.py     # evidence 1
.venv/bin/python proposal/scripts/capture_aws_config_evidence.py # evidence 2
.venv/bin/python proposal/scripts/capture_runtime_evidence.py  # evidence 3
.venv/bin/python proposal/scripts/capture_tests_evidence.py    # evidence 4
.venv/bin/python proposal/scripts/capture_dashboard_screenshot.py # evidence 5
.venv/bin/python proposal/scripts/capture_live_send.py --channel email --approve #
  evidence 6
.venv/bin/python proposal/scripts/capture_delivery_confirmation.py # evidence 7
.venv/bin/python proposal/scripts/capture_evidence_index.py     # index
.venv/bin/python proposal/scripts/build_pdf.py                 # this document
.venv/bin/python proposal/scripts/build_deployment_evidence_pdf.py # consolidated evidence
  pack
```

Evidence 6 transmits to a real mailbox and therefore refuses to run without `--approve` ; every other script is read-only with respect to the outside world. Evidence 1 depends on the generator in the first line, because the audit log is appended to rather than recomputed. Evidence 8 has no command here: its image was contributed by the operator rather than captured, which is why the index names it as hand-contributed instead of listing a script for it.

Three prerequisites apply. Evidences 2, 6 and 7 read live AWS state, so the shell must hold valid credentials for the account — on the machine used here that meant exporting them from the CLI session first, because the installed `boto3` predates the credential provider that would otherwise read them directly:

```
eval "$(aws configure export-credentials --format env | grep -v '^echo')"
```

Evidences 5 and 7 also drive a local browser and an IMAP client respectively (headless Chrome, and the mailbox credentials from `SMTP_USERNAME` / `SMTP_PASSWORD` in `.env`).

Each script writes both a human-readable report and the machine-readable facts behind it, plus the raw artifact where one exists, so a reviewer can recompute any figure in this document instead of trusting it.

`00_index.md` records the SHA-256 of every artifact at the moment the index was generated.